

D 3.1-14. Consider the following problem, where the values of c_1 and c_2 have not yet been ascertained.

$$\text{Maximize } Z = c_1x_1 + c_2x_2,$$

subject to

$$2x_1 + x_2 \leq 11$$

$$-x_1 + 2x_2 \leq 2$$

and

$$x_1 \geq 0, \quad x_2 \geq 0.$$

Use graphical analysis to determine the optimal solution(s) for (x_1, x_2) for the various possible values of c_1 and c_2 . (*Hint: Separate the cases where $c_2 = 0$, $c_2 > 0$, and $c_2 < 0$. For the latter two cases, focus on the ratio of c_1 to c_2 .)*)

5.1-8. Each of the following statements is true under most circumstances, but not always. In each case, indicate when the statement will not be true and why.

(a) The best CPF solution is an optimal solution.

(b) An optimal solution is a CPF solution.

(c) A CPF solution is the only optimal solution if none of its adjacent CPF solutions are better (as measured by the value of the objective function).

5.1-9. Consider the original form (before augmenting) of a linear programming problem with n decision variables (each with a nonnegativity constraint) and m functional constraints. Label each of the following statements as true or false, and then justify your answer with specific references (including page citations) to material in the chapter.

(a) If a feasible solution is optimal, it must be a CPF solution.

(b) The number of CPF solutions is at least $\binom{m+n}{n} = \frac{(m+n)!}{m!n!}$.

(c) If a CPF solution has adjacent CPF solutions that are better (as measured by Z), then one of these adjacent CPF solutions must be an optimal solution.

5.1-10. Label each of the following statements about linear programming problems as true or false, and then justify your answer.

(a) If a feasible solution is optimal but not a CPF solution, then infinitely many optimal solutions exist.

(b) If the value of the objective function is equal at two different feasible points $\mathbf{x}^*$ and $\mathbf{x}^{**}$, then all points on the line segment connecting $\mathbf{x}^*$ and $\mathbf{x}^{**}$ are feasible and Z has the same value at all those points.

(c) If the problem has n variables (before augmenting), then the simultaneous solution of any set of n constraint boundary equations is a CPF solution.

5.1-11. Consider the augmented form of linear programming problems that have feasible solutions and a bounded feasible region. Label each of the following statements as true or false, and then justify your answer by referring to specific statements (with page citations) in the chapter.

(a) There must be at least one optimal solution.

(b) An optimal solution must be a BF solution.

3.5-6. Maureen Laird is the chief financial officer for the Alva Electric Co., a major public utility in the midwest. The company has scheduled the construction of new hydroelectric plants 5, 10, and 20 years from now to meet the needs of the growing population in the region served by the company. To cover at least the construction costs, Maureen needs to invest some of the company's money now to meet these future cash-flow needs. Maureen may purchase only three kinds of financial assets, each of which costs \$1 million per unit. Fractional units may be purchased. The assets produce income 5, 10, and 20 years from now, and that income is needed to cover at least minimum cash-flow requirements in those years. (Any excess income above the minimum requirement for each time period will be used to increase dividend payments to shareholders rather than saving it to help meet the minimum cash-flow requirement in the next time period.) The following table shows both the amount of income generated by each unit of each asset and the minimum amount of income needed for each of the future time periods when a new hydroelectric plant will be constructed.

Year	Income per Unit of Asset			Minimum Cash Flow Required
	Asset 1	Asset 2	Asset 3	
5	\$2 million	\$1 million	\$0.5 million	\$400 million
10	\$0.5 million	\$0.5 million	\$1 million	\$100 million
20	0	\$1.5 million	\$2 million	\$300 million

Maureen wishes to determine the mix of investments in these assets that will cover the cash-flow requirements while minimizing the total amount invested.

- Formulate a linear programming model for this problem.
 - Display the model on a spreadsheet.
 - Use the spreadsheet to check the possibility of purchasing 100 units of Asset 1, 100 units of Asset 2, and 200 units of Asset 3. How much cash flow would this mix of investments generate 5, 10, and 20 years from now? What would be the total amount invested?
 - Take a few minutes to use a trial-and-error approach with the spreadsheet to develop your best guess for the optimal solution. What is the total amount invested for your solution?
- c (e) Use the Excel Solver to solve the model by the simplex method.

C 3.6-6. Reconsider Prob. 3.5-6. Use Gurobi to formulate and solve the model for this problem.

■ CASE 3.3 STAFFING A CALL CENTER¹

California Children's Hospital has been receiving numerous customer complaints because of its confusing, decentralized appointment and registration process. When customers want to make appointments or register child patients, they must contact the clinic or department they plan to visit. Several problems exist with this current strategy. Parents do not always know the most appropriate clinic or department they must visit to address their children's ailments. They therefore spend a significant amount of time on the phone being transferred from clinic to clinic until they reach the most appropriate clinic for their needs. The hospital also does not publish the phone numbers of all clinics and departments, and parents must therefore invest a large amount of time in detective work to track down the correct phone number. Finally, the various clinics and departments do not communicate with each other. For example, when a doctor schedules a referral with a colleague located in another department or clinic, that department or clinic almost never receives word of the referral. The parent must contact the correct department or clinic and provide the needed referral information.

In efforts to reengineer and improve its appointment and registration process, the children's hospital has decided to centralize the process by establishing one call center

devoted exclusively to appointments and registration. The hospital is currently in the middle of the planning stages for the call center. Jin Chin, the hospital manager, plans to operate the call center from 7 A.M. to 9 P.M. during the weekdays.

Several months ago, the hospital hired an ambitious management consulting firm, Creative Chaos Consultants, to forecast the number of calls the call center would receive each hour of the day. Since all appointment and registration-related calls would be received by the call center, the consultants decided that they could forecast the calls at the call center by totaling the number of appointment and registration-related calls received by all clinics and departments. The team members visited all the clinics and departments, where they diligently recorded every call relating to appointments and registration. They then totaled these calls and altered the totals to account for calls missed during data collection. They also altered totals to account for repeat calls that occurred when the same parent called the hospital many times because of the confusion surrounding the decentralized process. Creative Chaos Consultants determined the average number of calls the call center should expect during each hour of a weekday. The following table provides the forecasts.

Work Shift	Average Number of Calls
7 A.M.–9 A.M.	40 calls per hour
9 A.M.–11 A.M.	85 calls per hour
11 A.M.–1 P.M.	70 calls per hour
1 P.M.–3 P.M.	95 calls per hour
3 P.M.–5 P.M.	80 calls per hour
5 P.M.–7 P.M.	35 calls per hour
7 P.M.–9 P.M.	10 calls per hour

After the consultants submitted these forecasts, Jin became interested in the percentage of calls from Spanish speakers since the hospital services many Hispanic patients. Jin knows that he has to hire some operators who speak Spanish to handle these calls. The consultants performed further data collection and determined that on average, 20 percent of the calls were from Spanish speakers.

Given these call forecasts, Jin must now decide how to staff the call center during each 2-hour shift of a weekday. During the forecasting project, Creative Chaos Consultants closely observed the operators working at the individual clinics and departments and determined the number of calls operators process per hour. The consultants informed Jin that an operator is able to process an average of six calls per

For the following analysis consider only the labor cost for the time employees spend answering phones. The cost for paperwork time is charged to other cost centers.

- How many Spanish-speaking operators and how many English-speaking operators does the hospital need to staff the call center during each 2-hour shift of the day in order to answer all calls? Please provide an integer number since half a human operator makes no sense.
- Jin needs to determine how many full-time employees who speak Spanish, full-time employees who speak English, and part-time employees he should hire to begin on each shift. Creative Chaos Consultants advise him that linear programming can be used to do this in such a way as to minimize operating costs while answering all calls. Formulate a linear programming model of this problem.

hour. Jin also knows that he has both full-time and part-time workers available to staff the call center. A full-time employee works 8 hours per day, but because of paperwork that must also be completed, the employee spends only 4 hours per day on the phone. To balance the schedule, the employee alternates the 2-hour shifts between answering phones and completing paperwork. Full-time employees can start their day either by answering phones or by completing paperwork on the first shift. The full-time employees speak either Spanish or English, but none of them are bilingual. Both Spanish-speaking and English-speaking employees are paid \$20 per hour for work before 5 P.M. and \$24 per hour for work after 5 P.M. The full-time employees can begin work at the beginning of the 7 A.M. to 9 A.M. shift, 9 A.M. to 11 A.M. shift, 11 A.M. to 1 P.M. shift, or 1 P.M. to 3 P.M. shift. The part-time employees work for 4 hours, only answer calls, and only speak English. They can start work at the beginning of the 3 P.M. to 5 P.M. shift or the 5 P.M. to 7 P.M. shift, and like the full-time employees, they are paid \$20 per hour for work before 5 P.M. and \$24 per hour for work after 5 P.M.

- (c) Obtain an optimal solution for the linear programming model formulated in part (b) to guide Jin's decision.
- (d) Because many full-time workers do not want to work late into the evening, Jin can find only one qualified English-speaking operator willing to begin work at 1 P.M. Given this new constraint, how many full-time English-speaking operators, full-time Spanish-speaking operators, and part-time operators should Jin hire for each shift to minimize operating costs while answering all calls?
- (e) Jin now has decided to investigate the option of hiring bilingual operators instead of monolingual operators. If all the operators are bilingual, how many operators should be working during each 2-hour shift to answer all phone calls? As in part (a), please provide an integer answer.
- (f) If all employees are bilingual, how many full-time and part-time employees should Jin hire to begin on each shift to minimize operating costs while answering all calls? As in part (b), formulate a linear programming model to guide Jin's decision.
- (g) What is the maximum percentage increase in the hourly wage rate that Jin can pay bilingual employees over monolingual employees without increasing the total operating costs?
- (h) What other features of the call center should Jin explore to improve service or minimize operating costs?